Brittany Perera

brittanykawshalya@gmail.com

EXPERIENCE

AI Software Engineer May 2026 to now

PangeAI

Working with satellite images and extracting analysis from them for hazards, risks and changes to the earth over time.

Aviation Powerplant and Software Engineer (ML and predictive maintenance) Feb 2025 to May 2026

VietJet Air

- Built intelligent software into the airline dashboard for the Powerplant, Maintenance and Financial Management teams.
- Trained individual machine learning models for each engine in the fleet to predict deterioration, maintenance effectiveness, scheduling and shop visit optimisation.
- Built end-to-end data pipelines, feature engineering and simulation tools that forecast engine health.

Private Tutor Dec 2023 to May 2026

Oakley Tutors

- Taught A-level physics and maths part-time and remotely.

Undergraduate Researcher Oct 2023 to Jul 2024

Mercedes-AMG Petronas F1

- Optimised a radiator design for an F1 car using computational fluid dynamics and optimisation algorithms, as my master's project.

EDUCATION

MEng Engineering Science Oct 2020 to Jun 2024

University of Oxford, St Edmund Hall

Master's project: cooling unit optimisation for Mercedes-AMG Petronas F1. Key modules: Thermodynamics, Control Systems, Chemical Processes, Biomechanics, Biomedical Imaging.

A-levels in Maths, Physics and Chemistry (3 As) 2013 to 2020

Torquay Girls' Grammar School

LEADERSHIP

St Edmund Hall President Jan 2022 to Jan 2023

St Edmund Hall, Oxford

Represented 700+ students and led a committee of 15. Negotiated the student rent increase down from 13% to 8%.

Ball President Jun 2021 to Jun 2022

St Edmund Hall, Oxford

Ran an event for 900+ guests on a self-funded budget and managed contractors.

Summer Camp Counsellor Jun to Aug 2022

USA

Looked after children aged 6 to 17 with sports, woodwork, and camping and paddle-boarding expeditions.

SKILLS

Python, SQL, MATLAB, C++, TypeScript, JavaScript, SolidWorks, CAD, scikit-learn, GitHub Actions

LANGUAGES

English (native), Sinhala (fluent), French (intermediate), German (intermediate)